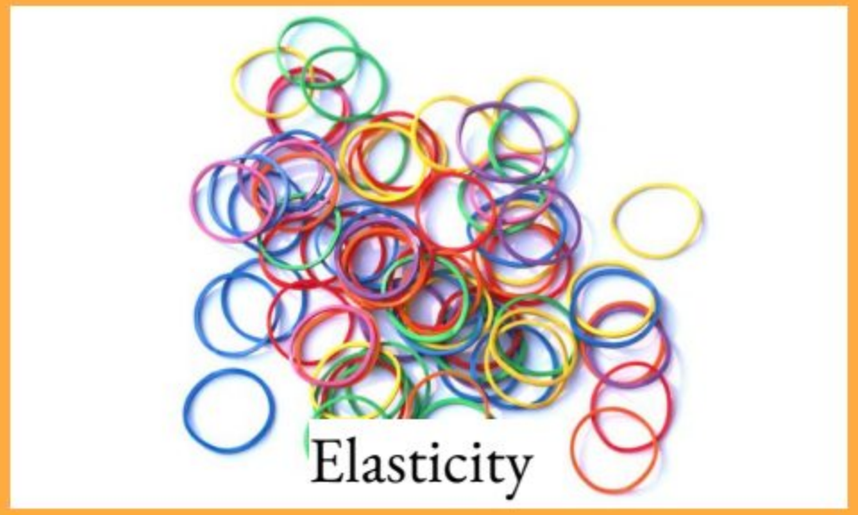
Properties and performance characteristics of material


Ductility

By the end of this you should be able to:

Clearly explain each term , give examples of materials that display the property in abundance , also show a material that is very poor in that property.


Malleability


Elasticity


Plasticity

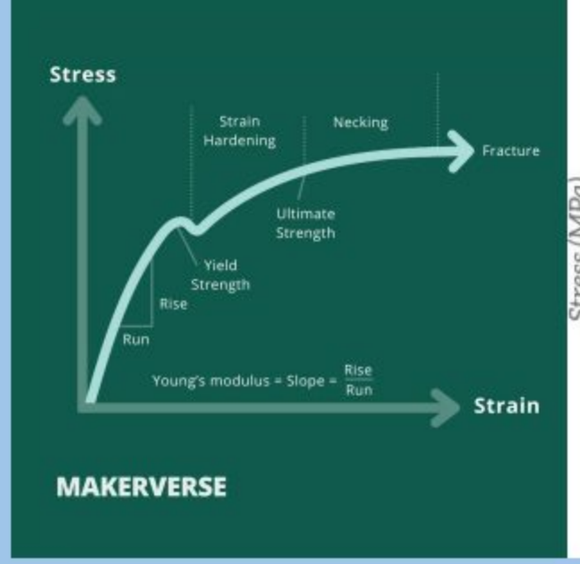
Ductility

Imagine stretching chewing gum — it thins out before snapping. That's ductility.

Ductility is a physical property of a material that describes its ability to be stretched, pulled, or drawn into a thin wire or elongated structure without breaking. It is a measure of how much plastic (permanent) deformation a material can sustain under tensile stress (pulling force) before it finally fractures.

Unlike **malleability**, which involves **compression (squeezing)**, ductility specifically refers to deformation under **tension** (stretching).

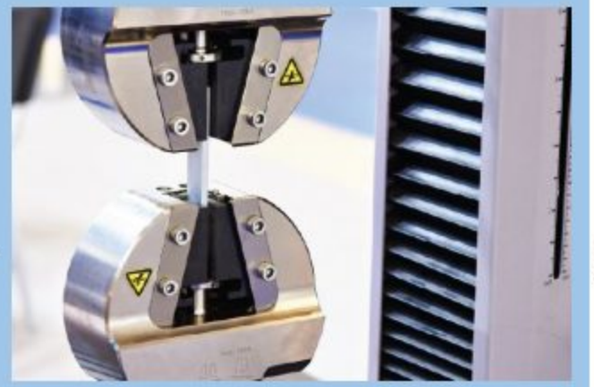
🧱 Brittle materials (like glass or cast iron) snap or shatter suddenly with little to no stretching. Ductile materials (like gold or copper) deform significantly before failing.

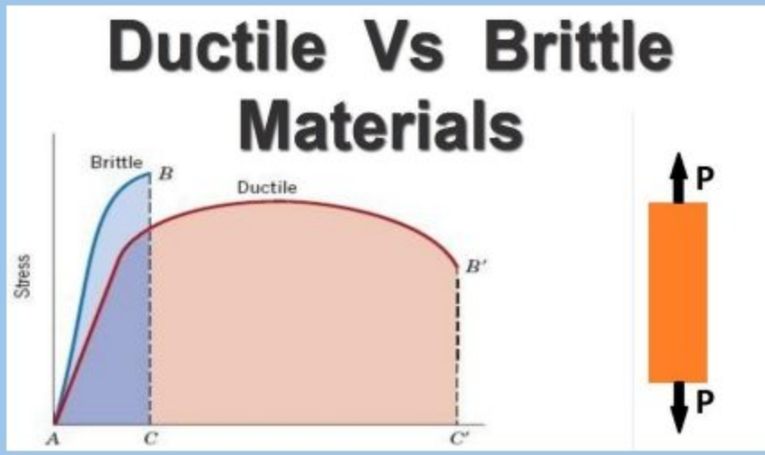




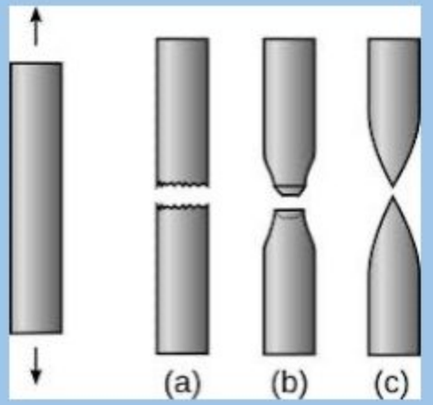
Factors Influencing Ductility

- **Temperature 🔥:** Most materials become more ductile at higher temperatures and more brittle at lower temperatures.
- **Alloying:** Adding other elements to a pure metal often decreases its ductility. For example, high-carbon steel is much more brittle than low-carbon steel.





HIGH	LOW
Copper wiring, gold jewellery, aluminium foil	Glass fibre, cast iron



A - Brittle fracture - ceramics and cold metals

B - Moderately ductile fracture, typical metals.

C - Very ductile, soft metals (e.g., Pb, Au) at room temperature, other metals, polymers, glasses at high temperatures.


σ
TOUGHNESS
STRENGTH
ε
DUCTILITY

Malleability 🔨



Malleability is a material's ability to deform under compressive stress (like hammering or rolling) into thin sheets or new shapes without fracturing, a key property of metals like gold, aluminum, and copper, crucial for manufacturing processes like stamping and forging.

Factors which affect malleability:

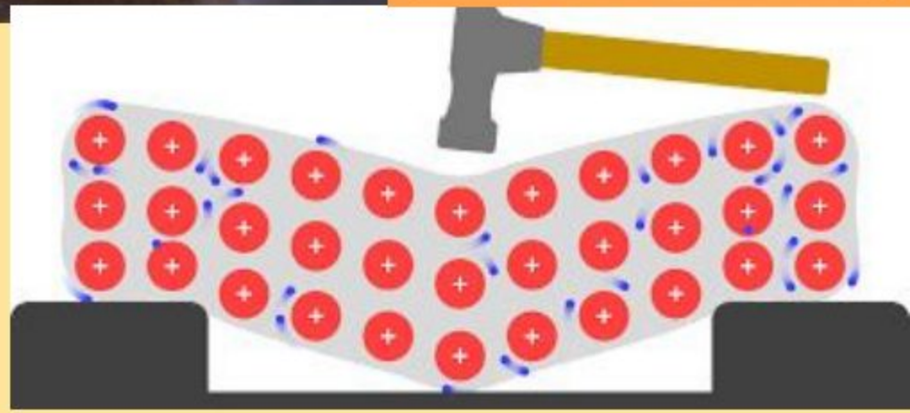
- **Temperature 🔥:** Most metals become **more malleable when heated** because the atoms gain energy and can move more freely.
- **Purity:** Pure metals are generally more malleable than alloys (mixtures of metals) because impurities or different-sized atoms in alloys can disrupt the smooth sliding of atomic layers.

HIGH	LOW
Gold leaf, aluminium cans	Zinc, cast iron

Think of flattening dough with a rolling pin — it spreads without cracking.







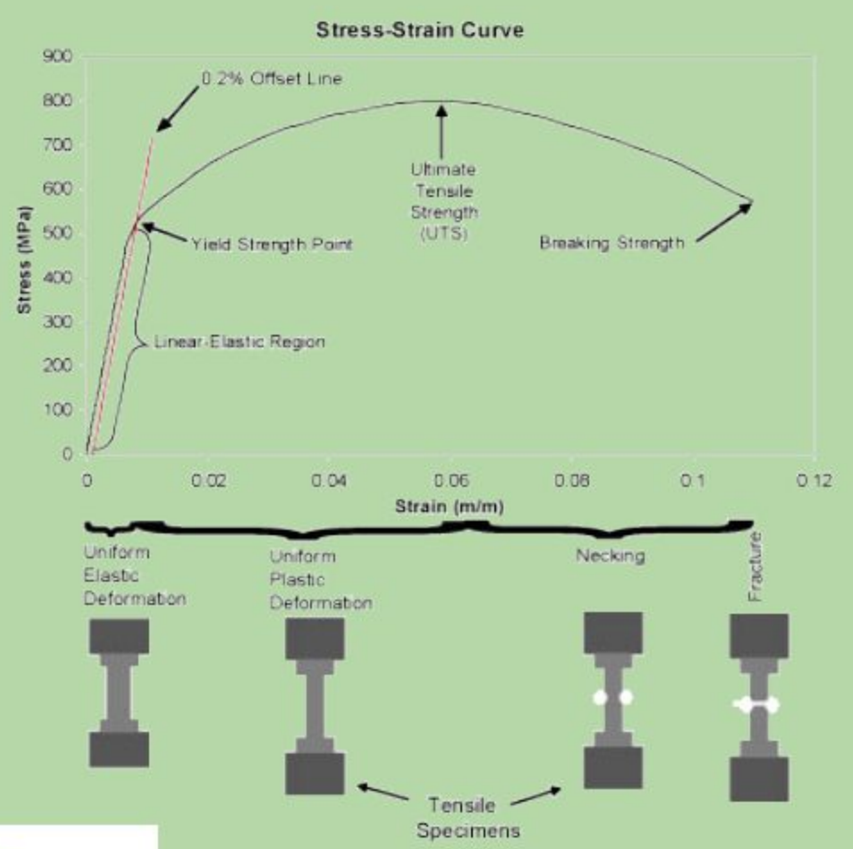
Elasticity 🌀

A trampoline stretches under you but springs back the moment you jump off.

Elasticity is a material's ability to deform under force but return to its original shape once the force is removed, like a stretched rubber band; it's governed by stress (force/area) causing strain (deformation), and described by Hooke's Law (stress proportional to strain within the elastic limit). Beyond the elastic limit, deformation becomes permanent (plasticity).

Factors which affect the elasticity of a material:

- Temperature 🔥: Generally, as temperature rises, elasticity decreases because molecular motion increases, making the material softer and less able to spring back.
- Impurities: Adding impurities can disrupt the uniform structure, either increasing elasticity (e.g., carbon in steel) or decreasing it, depending on the impurity.
- Processing and heat treatment:
    - Hammering/Rolling: Breaks crystals into smaller grains, increasing elasticity.
    - Annealing (slow heating/cooling): Creates larger, uniform grains, reducing elasticity and increasing softness.
    - Quenching (sudden cooling): Can make steel hard and less elastic.





Very elastic polymer

Not very elastic ceramic

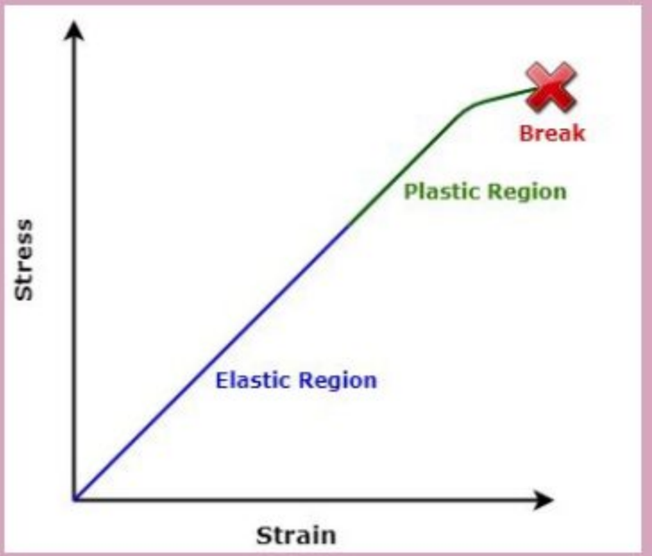


HIGH	LOW
Rubber bands, springs	Concrete

Plasticity

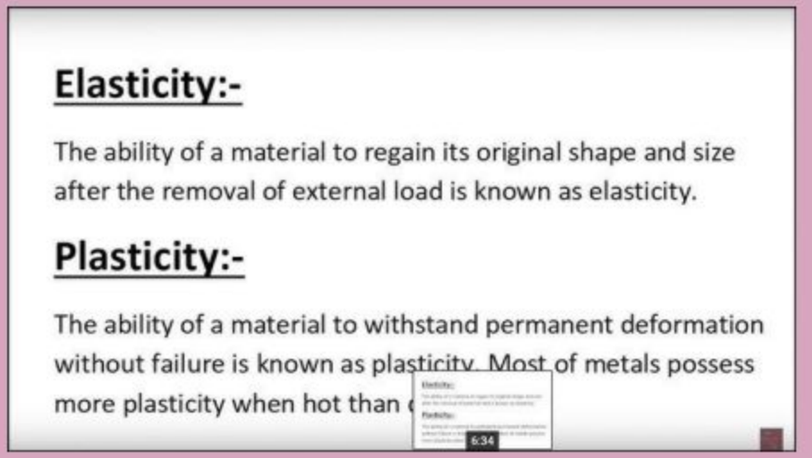
Why forging makes swords stronger.

Plasticity is a material's ability to deform permanently (change shape) without breaking when a force is applied, meaning it doesn't spring back to its original form like an elastic material does. It's crucial for shaping metals (like bending or forging) and seen in materials like clay, allowing for processes where a new, lasting shape is desired, contrasting with temporary elastic deformation.



Key aspects of plasticity:

- The deformation remains even after the stress is removed.
- It occurs after a material has passed its elastic limit (yield point) and entered the plastic region.
- examples: Bending a paperclip, shaping clay, or forging metal.



Elasticity is temporary (rubber band), while plasticity is permanent (bent metal).

HIGH	LOW
Hot steel, clay	Ceramics

Bend a paperclip once — it stays bent. That's plastic deformation.


Brittle
break
stress
strain